NBA Age vs. Performance: A Cross-Position Analysis

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1 Introduction

This report looks at how NBA player age relates to on-court performance across three position groups:Guards (Point Guards and Shooting Guards), Forwards (Power Forwards and Small Forwards),and Centers. We use data from the 2020–21 NBA season (Basketball Reference, 2021) and track six statistics: points, assists, offensive rebounds, defensive rebounds, field goal percentage, and turnovers.

Rather than using raw totals, we divide every stat by minutes played. This gives a per-minute rate that makes it fair to compare players who get very different amounts of playing time. The goal is to explore how these per-minute rates change with age for each position group.

2 Data and Methods

2.1 Data Source

The data comes from `nba-player-stats-2021.csv`, a publicly available dataset of NBA player statistics for the 2020–21 season (Basketball Reference, 2021). Each case is a single NBA player's season record. The attributes used are: `age` (years), `pos` (position: PG, SG, PF, SF, or C), `mp` (total minutes played), `pts` (points), `ast` (assists), `orb` (offensive rebounds), `drb` (defensive rebounds), `fgpercent` (field goal percentage, 0 to 1), and `tov` (turnovers). After normalization, all counting stats are rates per minute.

2.2 Player Selection Criteria

Players with fewer than 500 minutes played were excluded to avoid unreliable per-minute rates. Each position group is capped at the first 100 qualifying players.

2.3 FAIR and CARE Principles

The dataset satisfies the FAIR principles: it is publicly Findable and Accessible via Basketball Reference, stored in a standard CSV format (Interoperable), with clear column naming (Reusable). Applying the CARE principles is more challenging, as these guidelines target Indigenous data sovereignty. One challenge is that players had no input into how their statistics are collected or used in third-party analyses.

2.4 Per-Minute Normalization

Each counting stat (points, assists, rebounds, turnovers) and field goal percentage are divided by minutes played and rounded to six decimal places. The same process is applied consistently across all three position groups.

2.5 Analysis Paradigm

This analysis uses an Exploratory Data Analysis (EDA) approach (Tukey, 1977) — we use visualizations to look for patterns rather than test a specific hypothesis. All code is written using the Tidyverse (Wickham et al., 2019), chosen because the pipe operator and `ggplot2` produce readable, step-by-step code.

2.6 PCIP Plan Summary

The full PCIP plan is in Appendix 1. The four phases were: Plan (define the question and steps), Code (write the pipeline following the Tidyverse Style Guide), Implement (run and verify output), and Present (write the report with figures and appendices).

3 Guards: Point Guards and Shooting Guards

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$ g           <int> 73, 76, 56, 32, 47, 65, 50, 15, 66, 56, 54, 16, 3, 13, 69, ~
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$ fta         <dbl> 3.8, 4.8, 9.3, 4.3, 4.7, 3.7, 3.6, 4.0, 2.0, 6.4, 3.1, 3.0,~
$ ftpercent   <dbl> 0.595, 0.543, 0.753, 0.625, 0.873, 0.743, 0.722, 0.917, 0.8~
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```

Point Guards (PG) are mainly responsible for running the offense and setting up plays, while Shooting Guards (SG) focus more on scoring from the perimeter. Both spend most of their time on the outside of the court, so we would expect assists to be higher for PGs and scoring to be a priority for SGs. Figure Figure 1 shows how each per-minute stat looks across different ages for both guard types.

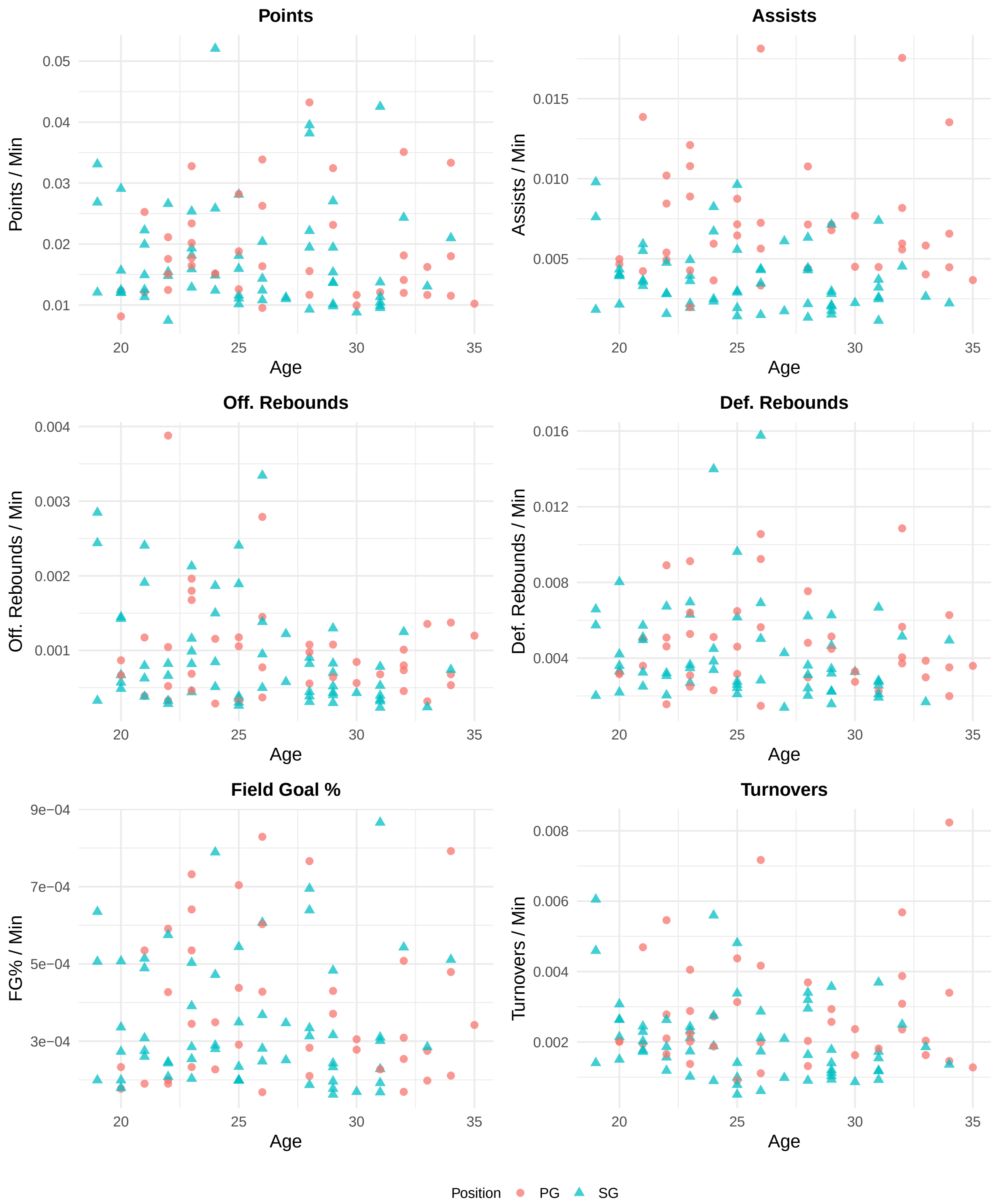

Figure 1: Age versus six per-minute performance metrics for NBA Guards (PG and SG) with at least 500 minutes played in the 2020–21 season. Each point represents one player, colored by position (PG = Point Guard, SG = Shooting Guard).

Looking at Figure Figure 1, midway through their twenties into their thirties, guards display varied scoring rates per minute, seen clearly in both point and shooting categories. While one might expect sharp differences, the data shows overlap instead. Notably, assists diverge - point guards consistently record higher numbers, matching expectations of them as primary creators. Unlike forwards, these backcourt players rarely collect offensive boards, given their positioning farther from the rim. Among younger cohorts, turnover counts scatter widely, yet they tighten with advancing age, perhaps due to greater experience shaping consistent responsibilities. Efficiency, measured by field goal accuracy, stays mostly flat across years for each group, without distinct peaks or declines. Table Table 1 shows mean and standard deviation for each per-minute stat across all 100 guards, and Table Table 2 lists the full player-level data.

3.1 Summary Statistics for Guards

Table 1: Mean and standard deviation of per-minute statistics for the 100 guards included in the analysis.

Variable	Mean	SD
age	26.0300	4.2958
pts	0.0182	0.0089
ast	0.0051	0.0033
orb	0.0009	0.0007
drb	0.0045	0.0026
fgpercent	0.0004	0.0002
tov	0.0024	0.0014

3.2 Table of Guards Used in Analysis

Table 2: Per-minute statistics for all 100 guards (PG and SG) included in the analysis.

player	pos	age	mp	pts	ast	orb	drb	fgpercent	tov
Nickeil Alexander-Walker	SG	23	1466	0.015894	0.003615	0.000819	0.003479	0.000254	0.002115
Nickeil Alexander-Walker	SG	23	1317	0.018147	0.003948	0.000987	0.003645	0.000285	0.002430
Grayson Allen	SG	26	1805	0.010803	0.001496	0.000499	0.002825	0.000248	0.000609
Jose Alvarado	PG	23	834	0.023381	0.010791	0.001799	0.005276	0.000535	0.002878
Cole Anthony	PG	21	2059	0.012093	0.004225	0.000389	0.003594	0.000190	0.001943
D.J. Augustin	PG	34	883	0.018007	0.006569	0.000680	0.003511	0.000479	0.003398
D.J. Augustin	PG	34	510	0.033333	0.013529	0.001373	0.006275	0.000792	0.008235
LaMelo Ball	PG	20	2422	0.012345	0.004666	0.000867	0.003220	0.000177	0.002023
Lonzo Ball	PG	24	1212	0.015099	0.005941	0.001155	0.005116	0.000349	0.002723
Desmond Bane	SG	23	2266	0.012886	0.001942	0.000441	0.002692	0.000203	0.001015
Dalano Banton	PG	22	696	0.021121	0.010201	0.003879	0.008908	0.000591	0.005460
Will Barton	SG	31	2277	0.009881	0.002591	0.000395	0.002811	0.000192	0.001186
Bradley Beal	SG	28	1439	0.022168	0.006324	0.000903	0.003614	0.000313	0.003197
Malik Beasley	SG	25	1976	0.011589	0.001417	0.000304	0.002429	0.000198	0.000506
Patrick Beverley	PG	33	1476	0.011653	0.005827	0.001355	0.003862	0.000275	0.001626
Eric Bledsoe	PG	32	1361	0.014107	0.005952	0.000735	0.004041	0.000309	0.003086
Bogdan Bogdanović	SG	29	1848	0.013690	0.002814	0.000433	0.003193	0.000233	0.001028
Devin Booker	SG	25	2345	0.015949	0.002900	0.000384	0.002601	0.000199	0.001407
Brandon Boston Jr.	SG	20	760	0.029079	0.004342	0.001447	0.008026	0.000507	0.002632
Avery Bradley	SG	31	1406	0.009531	0.001138	0.000782	0.002560	0.000301	0.000925
Malcolm Brogdon	PG	29	1206	0.023134	0.007131	0.001078	0.005141	0.000371	0.002570
Armoni Brooks	SG	23	690	0.025362	0.004928	0.001159	0.006957	0.000503	0.002319
Sterling Brown	SG	26	628	0.020382	0.004299	0.003344	0.015764	0.000607	0.002866

Jalen Brunson	SG	25	2524	0.010143	0.002971	0.000357	0.002100	0.000199	0.000990
Alec Burks	SG	30	2318	0.008801	0.002243	0.000431	0.003279	0.000169	0.000863
Kentavious Caldwell-Pope	SG	28	2329	0.009274	0.001331	0.000386	0.002018	0.000187	0.000902
Facundo Campazzo	PG	30	1184	0.011655	0.007686	0.000845	0.003294	0.000305	0.002365
Jevon Carter	PG	26	905	0.016354	0.005635	0.000773	0.005635	0.000428	0.001989
Jevon Carter	PG	26	552	0.026268	0.007246	0.001449	0.009239	0.000603	0.004167
Alex Caruso	SG	27	1147	0.011247	0.006103	0.001221	0.004272	0.000347	0.002092
Josh Christopher	SG	20	1334	0.015667	0.004048	0.001424	0.003598	0.000336	0.003073
Jordan Clarkson	SG	29	2141	0.013638	0.002102	0.000701	0.002242	0.000196	0.001401
Amir Coffey	SG	24	1567	0.012380	0.002489	0.000511	0.003382	0.000289	0.000893
Mike Conley	PG	34	2058	0.011516	0.004470	0.000534	0.001992	0.000211	0.001458
Cade Cunningham	SG	20	2088	0.012452	0.003975	0.000670	0.003305	0.000199	0.002634
Seth Curry	SG	31	2135	0.010445	0.002482	0.000234	0.001920	0.000228	0.001171
Seth Curry	SG	31	1566	0.013729	0.003704	0.000319	0.002746	0.000310	0.001724
Seth Curry	SG	31	569	0.042531	0.007381	0.000527	0.006678	0.000866	0.003691
Stephen Curry	PG	33	2211	0.016237	0.004025	0.000317	0.002985	0.000198	0.002035
Terence Davis	SG	24	536	0.052052	0.006716	0.001866	0.013993	0.000789	0.005597
Hamidou Diallo	SG	23	1269	0.019307	0.002206	0.002128	0.006304	0.000391	0.001734
Spencer Dinwiddie	PG	28	1980	0.011667	0.004394	0.000556	0.002980	0.000210	0.001313
Spencer Dinwiddie	PG	28	1330	0.015564	0.007143	0.000977	0.004812	0.000283	0.002030
Spencer Dinwiddie	PG	28	650	0.043231	0.010769	0.001077	0.007538	0.000766	0.003692
Donte DiVincenzo	SG	25	1006	0.018091	0.005567	0.001889	0.006163	0.000349	0.003380
Donte DiVincenzo	SG	25	665	0.028120	0.009624	0.002406	0.009624	0.000544	0.004812
Luka Dončić	PG	22	2301	0.017558	0.005389	0.000522	0.005085	0.000199	0.002781
Luguentz Dort	SG	22	1665	0.015435	0.001562	0.000661	0.003063	0.000243	0.001562
Ayo Dosunmu	SG	22	2110	0.007441	0.002796	0.000284	0.002038	0.000246	0.001185
Chris Duarte	SG	24	1541	0.014860	0.002336	0.000844	0.003829	0.000280	0.001882
Anthony Edwards	SG	20	2466	0.012003	0.002149	0.000487	0.002190	0.000179	0.001500
Wayne Ellington	SG	34	810	0.020988	0.002222	0.000741	0.004938	0.000511	0.001358
Malachi Flynn	PG	23	537	0.032775	0.012104	0.001676	0.009125	0.000732	0.002235
Bryn Forbes	SG	28	1286	0.019440	0.002177	0.000311	0.002411	0.000334	0.001633
Bryn Forbes	SG	28	676	0.038166	0.004290	0.000444	0.006213	0.000639	0.003402
Bryn Forbes	SG	28	610	0.039508	0.004426	0.000820	0.003115	0.000695	0.002951
Trent Forrest	PG	23	765	0.016471	0.008889	0.001961	0.006405	0.000641	0.004052
Evan Fournier	SG	29	2358	0.010136	0.001527	0.000297	0.001569	0.000177	0.000933
De'Aaron Fox	PG	24	2083	0.015170	0.003649	0.000288	0.002304	0.000227	0.001872
Darius Garland	PG	22	2430	0.012469	0.004938	0.000329	0.001564	0.000190	0.002099
Shai Gilgeous-Alexander	PG	23	1942	0.017714	0.004274	0.000463	0.003090	0.000233	0.002008
Brandon Goodwin	PG	26	502	0.033865	0.018127	0.002789	0.010558	0.000829	0.007171
Eric Gordon	SG	33	1669	0.013062	0.002636	0.000240	0.001678	0.000285	0.001857
Devonte' Graham	PG	26	2162	0.009528	0.003330	0.000370	0.001480	0.000168	0.001110
Jalen Green	SG	19	2138	0.012067	0.001824	0.000327	0.002011	0.000199	0.001403
Josh Green	SG	21	1039	0.014918	0.003657	0.002406	0.005101	0.000489	0.002021
Quentin Grimes	SG	21	786	0.022265	0.003562	0.001908	0.005725	0.000514	0.002290
Tyrese Haliburton	SG	21	1757	0.011326	0.005919	0.000626	0.002504	0.000260	0.001764
Tyrese Haliburton	PG	21	938	0.025267	0.013859	0.001173	0.005011	0.000535	0.004691
R.J. Hampton	SG	20	1402	0.012054	0.003923	0.000571	0.004208	0.000273	0.002140
Tim Hardaway Jr.	SG	29	1245	0.019438	0.002972	0.000402	0.004659	0.000316	0.001124
James Harden	PG	32	2419	0.011988	0.005581	0.000455	0.003721	0.000169	0.002356
James Harden	PG	32	1627	0.018132	0.008175	0.000799	0.005655	0.000254	0.003872
James Harden	PG	32	792	0.035101	0.017551	0.001010	0.010859	0.000508	0.005682
Gary Harris	SG	27	1730	0.010983	0.001734	0.000578	0.001387	0.000251	0.000983
Josh Hart	SG	26	1791	0.012395	0.003462	0.000949	0.005025	0.000281	0.001731
Josh Hart	SG	26	1374	0.014338	0.004367	0.001383	0.006914	0.000368	0.002111
Killian Hayes	PG	20	1647	0.008136	0.004979	0.000668	0.003157	0.000233	0.002004
Tyler Herro	SG	22	2151	0.014784	0.002836	0.000325	0.003208	0.000208	0.001860
Buddy Hield	SG	29	2499	0.009804	0.001761	0.000520	0.002241	0.000162	0.001200
Buddy Hield	SG	29	1574	0.015375	0.002033	0.000826	0.003431	0.000243	0.001779
Buddy Hield	SG	29	925	0.027027	0.007135	0.001297	0.006270	0.000483	0.003568
George Hill	PG	35	1253	0.010215	0.003671	0.001197	0.003591	0.000342	0.001277
Aaron Holiday	PG	25	1021	0.018805	0.007150	0.001175	0.004603	0.000438	0.003134
Aaron Holiday	PG	25	663	0.028205	0.008748	0.001056	0.006486	0.000704	0.004374
Jrue Holiday	PG	31	2207	0.012098	0.004486	0.000680	0.002266	0.000227	0.001812
Justin Holiday	SG	32	641	0.024337	0.004524	0.001248	0.005148	0.000543	0.002496

Talen Horton-Tucker	SG	21	1511	0.012508	0.003309	0.000794	0.003243	0.000275	0.001721
Bones Hyland	SG	21	1310	0.019924	0.005496	0.000382	0.004962	0.000308	0.002443
Kyrie Irving	PG	29	1091	0.032447	0.006783	0.000642	0.004491	0.000430	0.002933
Frank Jackson	PG	23	1164	0.020189	0.001976	0.000687	0.002491	0.000345	0.001375
Reggie Jackson	SG	31	2337	0.011339	0.003209	0.000342	0.002097	0.000168	0.001540
Ty Jerome	SG	24	801	0.025843	0.008240	0.001498	0.004494	0.000472	0.002747
Isaiah Joe	SG	22	609	0.026601	0.004762	0.000821	0.006732	0.000575	0.002627
Keon Johnson	SG	19	697	0.026829	0.007604	0.002439	0.005739	0.000506	0.004591
Keon Johnson	SG	19	562	0.033096	0.009786	0.002847	0.006584	0.000635	0.006050
Tre Jones	PG	22	1148	0.015070	0.008449	0.001045	0.004617	0.000427	0.001655
Tyus Jones	PG	25	1549	0.012589	0.006456	0.000323	0.003163	0.000291	0.000904
Cory Joseph	PG	30	1600	0.009938	0.004500	0.000563	0.002750	0.000278	0.001625
Luke Kennard	SG	25	1919	0.011100	0.001928	0.000261	0.002762	0.000234	0.000782

4 Forwards: Power Forwards and Small Forwards

Power Forwards (PF) tend to play closer to the basket and focus on rebounding and inside scoring. Small Forwards (SF) are more versatile — they can score from the perimeter and also get to the rim. Because these two roles use the court differently, it is interesting to see how age affects each one. Figure Figure 2 shows the per-minute stats for both forward types.

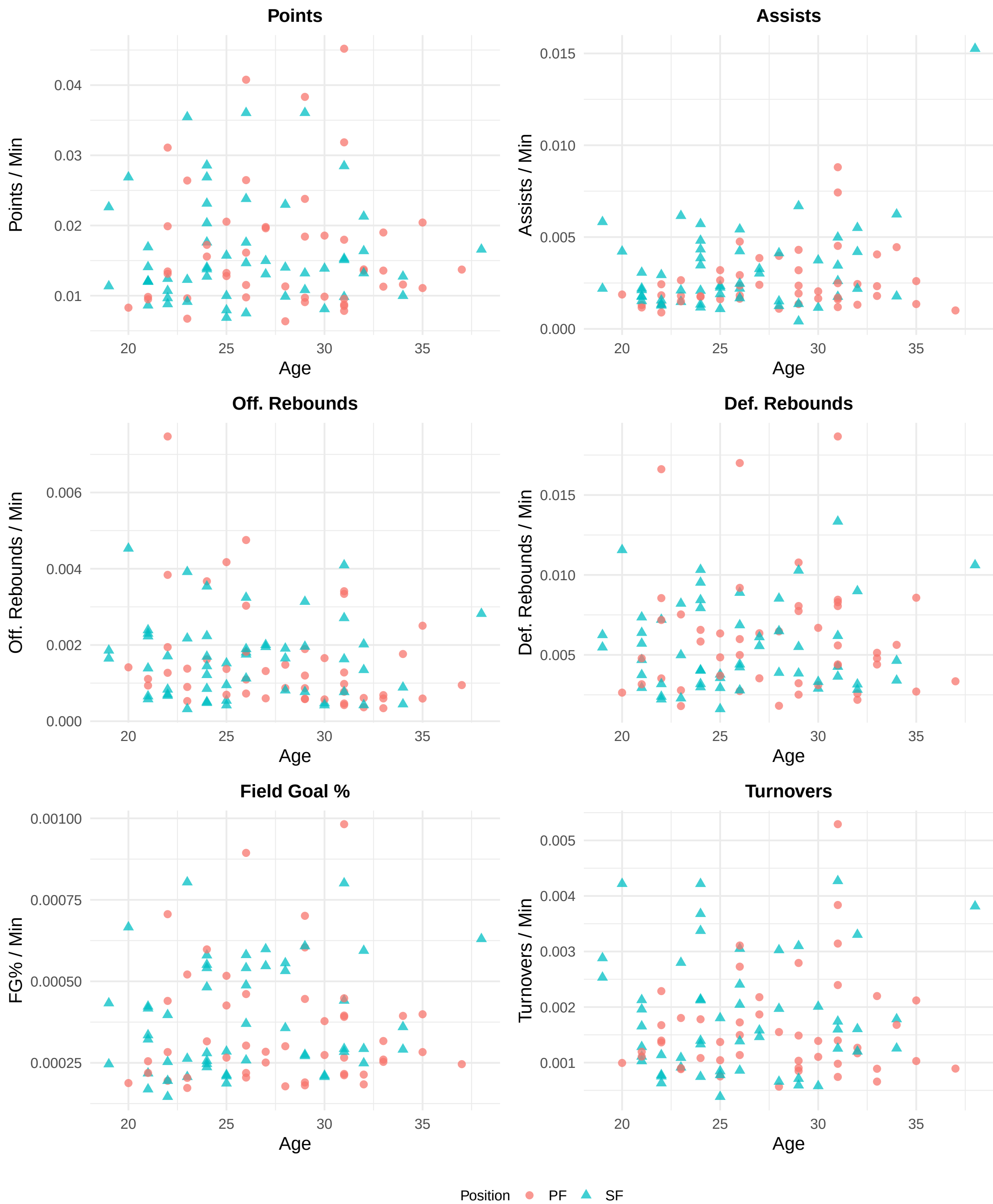

Figure 2: Age versus six per-minute performance metrics for NBA Forwards (PF and SF) with at least 500 minutes played in the 2020–21 season. Each point represents one player, colored by position (PF = Power Forward, SF = Small Forward).

Figure Figure 2 shows that Scoring each minute holds steady through the mid-twenties, though a bit more often peaks between ages twenty-six and thirty. While assists generally run lower for forwards compared to guards, several small forwards past age twenty-five post notably higher figures. What stands apart is rebounding - power forwards grab boards far more frequently than small forwards, regardless of age, without narrowing much over time. That pattern hints role shapes rebound output stronger than aging does. Younger players show wider turnover swings early on; these tend to even out later, echoing trends seen earlier in guard data. Table Table 3 summarizes the per-minute statistics and Table Table 4 lists all 100 players.

4.1 Summary Statistics for Forwards

Table 3: Mean and standard deviation of per-minute statistics for the 100 forwards included in the analysis.

Variable	Mean	SD
age	26.8200	4.3955
pts	0.0162	0.0081
ast	0.0028	0.0020
orb	0.0016	0.0012
drb	0.0057	0.0033
fgpercent	0.0004	0.0002
tov	0.0017	0.0010

4.2 Table of Forwards Used in Analysis

Table 4: Per-minute statistics for all 100 forwards (PF and SF) included in the analysis.

player	pos	age	mp	pts	ast	orb	drb	fgpercent	tov
Kyle Anderson	PF	28	1484	0.011321	0.003976	0.001482	0.006469	0.000301	0.001550
Giannis Antetokounmpo	PF	27	2204	0.019782	0.003857	0.001316	0.006352	0.000251	0.002178
Carmelo Anthony	PF	37	1793	0.013720	0.001004	0.000948	0.003346	0.000246	0.000892
OG Anunoby	SF	24	1728	0.013773	0.002083	0.001215	0.003183	0.000256	0.001331
Deni Avdija	SF	21	1984	0.008619	0.002117	0.000655	0.004688	0.000218	0.001109
Marvin Bagley III	PF	22	1146	0.019895	0.001309	0.003839	0.008551	0.000440	0.001396
Marvin Bagley III	PF	22	656	0.031098	0.001829	0.007470	0.016616	0.000706	0.002287
Harrison Barnes	PF	29	2587	0.009084	0.001353	0.000580	0.002513	0.000181	0.000850
Scottie Barnes	PF	20	2617	0.008292	0.001872	0.001414	0.002637	0.000188	0.000994
RJ Barrett	SF	21	2417	0.011998	0.001779	0.000579	0.002938	0.000169	0.001283
Keita Bates-Diop	SF	26	956	0.017573	0.002197	0.003243	0.008891	0.000541	0.002406
Nicolas Batum	PF	33	1462	0.011286	0.002326	0.000684	0.005130	0.000317	0.000889
Kent Bazemore	SF	32	545	0.021284	0.005505	0.002018	0.008991	0.000594	0.003303
Darius Bazley	PF	21	1924	0.009823	0.001299	0.000936	0.004782	0.000219	0.001195
DeAndre' Bembry	SF	27	1026	0.013060	0.003021	0.001949	0.005556	0.000547	0.001462
DeAndre' Bembry	SF	27	949	0.014963	0.003267	0.002002	0.006112	0.000599	0.001581
Dāvis Bertāns	PF	29	807	0.023792	0.002354	0.000867	0.008055	0.000446	0.001487
Dāvis Bertāns	PF	29	501	0.038323	0.003194	0.001198	0.010778	0.000701	0.002794
Saddiq Bey	SF	22	2704	0.008802	0.001553	0.000703	0.002219	0.000146	0.000629
Bojan Bogdanović	PF	32	2131	0.013562	0.001314	0.000610	0.002581	0.000214	0.001267
Chris Boucher	SF	29	1690	0.013195	0.000414	0.003136	0.005503	0.000275	0.000710
Ignas Brazdeikis	SF	23	536	0.035448	0.006157	0.003918	0.008209	0.000804	0.002799
Mikal Bridges	SF	25	2854	0.006868	0.001086	0.000420	0.001612	0.000187	0.000385
Miles Bridges	PF	23	2837	0.009623	0.001798	0.000529	0.002785	0.000173	0.000881
Oshae Brissett	SF	23	1564	0.012276	0.001471	0.002174	0.004987	0.000263	0.001087
Dillon Brooks	SF	26	885	0.036045	0.005424	0.001808	0.004407	0.000488	0.003051

Bruce Brown	SF	25	1774	0.009977	0.002255	0.001522	0.003777	0.000285	0.000846
Greg Brown III	SF	20	640	0.026875	0.004219	0.004531	0.011562	0.000666	0.004219
Jaylen Brown	SF	25	2220	0.015721	0.002342	0.000541	0.003559	0.000213	0.001802
Troy Brown Jr.	SF	22	1055	0.012417	0.002938	0.001706	0.007204	0.000397	0.001137
Reggie Bullock	SF	30	1902	0.008097	0.001157	0.000473	0.002892	0.000211	0.000578
Jimmy Butler	SF	32	1931	0.016365	0.004195	0.001346	0.003159	0.000249	0.001605
Wendell Carter Jr.	PF	22	1852	0.013121	0.002430	0.001944	0.007181	0.000283	0.001674
Brandon Clarke	PF	25	1246	0.020546	0.002648	0.004173	0.006340	0.000517	0.001043
John Collins	PF	24	1663	0.015574	0.001744	0.001624	0.005833	0.000316	0.001082
Pat Connaughton	SF	29	1691	0.010822	0.001360	0.000769	0.003844	0.000271	0.000591
Robert Covington	PF	31	1940	0.007835	0.001186	0.000773	0.004330	0.000216	0.000979
Robert Covington	PF	31	1431	0.008735	0.001607	0.000978	0.005590	0.000266	0.001398
Robert Covington	PF	31	509	0.045187	0.004519	0.003340	0.018664	0.000982	0.003143
Torrey Craig	SF	31	1596	0.009837	0.001754	0.001629	0.004261	0.000284	0.001253
Torrey Craig	SF	31	1034	0.015087	0.002611	0.002708	0.006190	0.000441	0.001741
Torrey Craig	SF	31	562	0.028470	0.004982	0.004093	0.013345	0.000801	0.004270
Jae Crowder	PF	31	1886	0.008537	0.001697	0.000424	0.004401	0.000212	0.000742
DeMar DeRozan	PF	32	2743	0.013744	0.002443	0.000365	0.002187	0.000184	0.001167
Kevin Durant	PF	33	2047	0.019003	0.004055	0.000342	0.004397	0.000253	0.002198
Kessler Edwards	SF	21	987	0.014083	0.001520	0.002229	0.006383	0.000417	0.002128
CJ Elleby	SF	21	1174	0.012010	0.003066	0.002300	0.005707	0.000335	0.001959
Dorian Finney-Smith	PF	28	2644	0.006354	0.001097	0.000870	0.001815	0.000178	0.000567
Danilo Gallinari	PF	33	1672	0.013577	0.001794	0.000598	0.004785	0.000260	0.000658
Rudy Gay	PF	35	1038	0.020424	0.002601	0.002505	0.008574	0.000399	0.002119
Paul George	PF	31	1077	0.031848	0.007428	0.000464	0.008449	0.000391	0.005292
Josh Giddey	SF	19	1700	0.011353	0.005824	0.001647	0.005471	0.000246	0.002882
Aaron Gordon	PF	26	2376	0.009764	0.001641	0.001094	0.002736	0.000219	0.001136
Jerami Grant	PF	27	1500	0.019600	0.002400	0.000600	0.003533	0.000284	0.001867
Danny Green	SF	34	1353	0.009978	0.001774	0.000887	0.003400	0.000291	0.001256
Draymond Green	PF	31	1329	0.009556	0.008804	0.001279	0.008051	0.000395	0.003837
JaMychal Green	PF	31	1085	0.017972	0.002488	0.003410	0.008295	0.000448	0.002396
Javonte Green	SF	28	1519	0.009875	0.001251	0.001909	0.003884	0.000357	0.000658
Jeff Green	PF	35	1849	0.011087	0.001352	0.000595	0.002704	0.000283	0.001028
Rui Hachimura	PF	23	943	0.026405	0.002651	0.001379	0.007529	0.000521	0.001803
Maurice Harkless	SF	28	863	0.014021	0.001506	0.000811	0.006489	0.000532	0.001970
Tobias Harris	PF	29	2543	0.009713	0.001927	0.000590	0.003225	0.000190	0.000904
Gordon Hayward	SF	31	1564	0.015281	0.003453	0.000767	0.003645	0.000293	0.001598
Justin Holiday	SF	32	1416	0.013206	0.002189	0.000424	0.002825	0.000293	0.001201
Rodney Hood	PF	29	581	0.018417	0.004303	0.001893	0.007745	0.000604	0.001033
Danuel House Jr.	SF	28	727	0.022971	0.004127	0.001651	0.008528	0.000556	0.003026
Kevin Huerter	SF	23	2188	0.009141	0.002102	0.000320	0.002285	0.000207	0.000914
De'Andre Hunter	SF	24	1577	0.014014	0.001332	0.000507	0.002980	0.000280	0.001395
Andre Iguodala	SF	38	603	0.016584	0.015257	0.002819	0.010614	0.000630	0.003814
Joe Ingles	SF	34	1122	0.012745	0.006239	0.000446	0.004635	0.000360	0.001783
Brandon Ingram	SF	24	1869	0.017603	0.004334	0.000482	0.004013	0.000247	0.002140
Jaren Jackson Jr.	PF	22	2126	0.013452	0.000894	0.001270	0.003528	0.000195	0.001364
Josh Jackson	SF	24	830	0.023133	0.003855	0.001446	0.008434	0.000482	0.003373
Josh Jackson	SF	24	707	0.026874	0.004809	0.001697	0.010325	0.000580	0.003678
Cameron Johnson	PF	25	1730	0.013237	0.001618	0.000694	0.003699	0.000266	0.000751
James Johnson	PF	34	1191	0.011587	0.004450	0.001763	0.005626	0.000394	0.001679
Keldon Johnson	SF	22	2392	0.010702	0.001338	0.000669	0.003177	0.000195	0.000753
Stanley Johnson	PF	25	1094	0.012797	0.003199	0.001371	0.004845	0.000426	0.001371
Derrick Jones Jr.	PF	24	899	0.017241	0.001780	0.003671	0.006563	0.000598	0.001780
Herbert Jones	PF	23	2335	0.006724	0.001499	0.000899	0.001799	0.000204	0.000899
Corey Kispert	SF	22	1801	0.009661	0.001277	0.000833	0.002388	0.000253	0.000777
Maxi Kleber	PF	30	1450	0.009862	0.001655	0.001655	0.006690	0.000274	0.001103
Jonathan Kuminga	SF	19	1185	0.022616	0.002194	0.001857	0.006245	0.000433	0.002532
Kyle Kuzma	PF	26	2204	0.011525	0.002359	0.000726	0.004991	0.000205	0.001724
Jeremy Lamb	SF	29	613	0.036052	0.006688	0.001958	0.010277	0.000608	0.003100
Nassir Little	SF	21	1088	0.016912	0.002206	0.002390	0.007353	0.000423	0.001654
Timothé Luwawu-Cabarrot	SF	26	685	0.023796	0.004234	0.001752	0.006861	0.000581	0.002044
Trey Lyles	PF	26	1537	0.016135	0.001822	0.001822	0.005986	0.000303	0.001496
Trey Lyles	PF	26	990	0.026465	0.002929	0.003030	0.009192	0.000461	0.002727
Trey Lyles	PF	26	547	0.040768	0.004753	0.004753	0.017002	0.000894	0.003108

Terance Mann	SF	25	2317	0.007941	0.001899	0.000950	0.002935	0.000209	0.000777
Lauri Markkanen	SF	24	1878	0.012726	0.001171	0.000852	0.004047	0.000237	0.000745
Naji Marshall	SF	24	735	0.028571	0.005714	0.003537	0.009524	0.000551	0.004218
Caleb Martin	SF	26	1372	0.014650	0.001676	0.001895	0.004227	0.000370	0.001385
Cody Martin	SF	26	1866	0.007503	0.002465	0.001125	0.002787	0.000258	0.000857
Kenyon Martin Jr.	SF	21	1656	0.012077	0.001751	0.001389	0.003744	0.000322	0.001027
Jaden McDaniels	PF	21	1803	0.009484	0.001165	0.001109	0.003161	0.000255	0.001109
Jalen McDaniels	SF	24	895	0.020335	0.003464	0.002235	0.007933	0.000541	0.002123
Doug McDermott	PF	30	1223	0.018561	0.002044	0.000572	0.003107	0.000378	0.001390
Khris Middleton	SF	30	2141	0.013872	0.003737	0.000420	0.003316	0.000207	0.002008

5 Centers

Centers generally play near the basket. Their main jobs are rebounding, protecting the rim, and scoring inside. Because there is only one position label (C), all points in Figure Figure 3 are the same color, so the focus is entirely on how age relates to each stat.

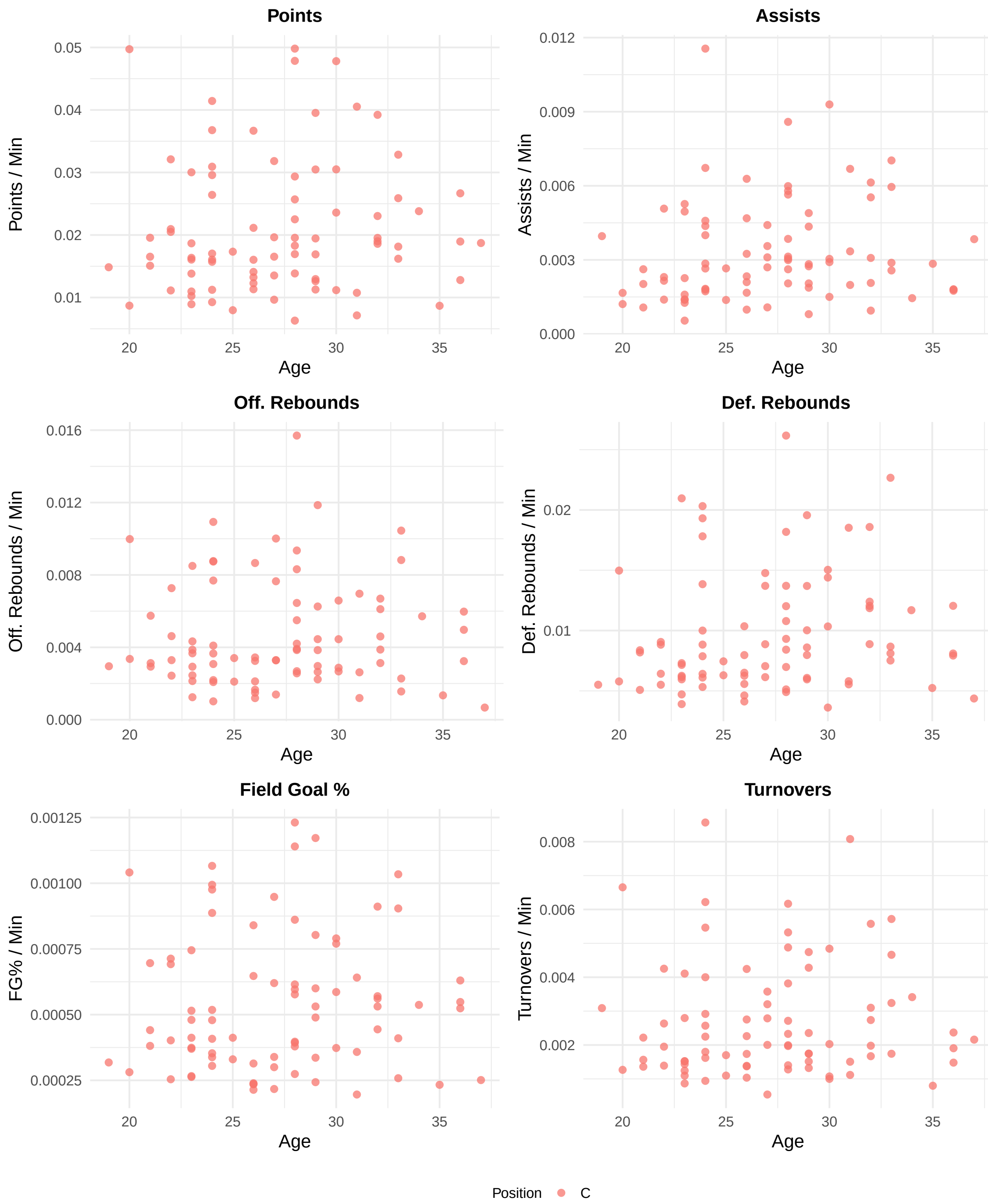

Figure 3: Age versus six per-minute performance metrics for NBA Centers (C) with at least 500 minutes played in the 2020–21 season. Each point represents one player. All players share the same position label (C).

Looking at Figure Figure 3, scoring rates for centers stay close together - most fall from 0.4 to 0.8 points each minute - with little shift across ages. Because creating shots isn't typically their role, it follows that assist numbers sit below those of other positions. Unlike guards or forwards, these players pull down more rebounds, as seen when comparing with Figures Figure 1 and Figure 2; yet a small decline appears in defensive boards past the early years.

By the third decade of life, mobility tends to decrease slightly. Although turnover rates begin elevated among new center players, they tend to decline with experience. This trend mirrors what is observed in back court and wing positions alike. Table Table 5 provides summary statistics and Table Table 6 lists all centers used.

5.1 Summary Statistics for Centers

Table 5: Mean and standard deviation of per-minute statistics for the centers included in the analysis.

Variable	Mean	SD
age	27.1975	4.2527
pts	0.0209	0.0107
ast	0.0033	0.0021
orb	0.0046	0.0030
drb	0.0097	0.0050
fgpercent	0.0005	0.0003
tov	0.0027	0.0017

5.2 Table of Centers Used in Analysis

Table 6: Per-minute statistics for all centers (C) included in the analysis.

player	pos	age	mp	pts	ast	orb	drb	fgpercent	tov
Precious Achiuwa	C	22	1725	0.011130	0.001391	0.002435	0.005507	0.000254	0.001391
Steven Adams	C	28	1999	0.006303	0.003052	0.004202	0.004902	0.000274	0.001401
Bam Adebayo	C	24	1825	0.016055	0.002849	0.002082	0.006411	0.000305	0.002247
LaMarcus Aldridge	C	36	1050	0.026667	0.001810	0.003238	0.008095	0.000524	0.001905
Jarrett Allen	C	23	1809	0.013820	0.001382	0.002930	0.006247	0.000374	0.001437
Deandre Ayton	C	23	1713	0.016346	0.001401	0.002452	0.007297	0.000370	0.001518
Mo Bamba	C	23	1824	0.010965	0.001261	0.002138	0.006140	0.000263	0.001096
Khem Birch	C	29	991	0.012614	0.002825	0.006256	0.005954	0.000489	0.001514
Goga Bitadze	C	22	729	0.032099	0.005075	0.007270	0.009053	0.000713	0.004252
Bismack Biyombo	C	29	506	0.039526	0.004348	0.011858	0.019565	0.001172	0.004743
Nemanja Bjelica	C	33	1142	0.016200	0.005954	0.002277	0.008669	0.000410	0.003240
Tony Bradley	C	24	549	0.026412	0.004372	0.010929	0.019308	0.001066	0.005464
Clint Capela	C	27	2042	0.009647	0.001077	0.003281	0.007052	0.000300	0.000539
Nic Claxton	C	22	974	0.020945	0.002156	0.004620	0.008830	0.000692	0.001951
Zach Collins	C	24	502	0.041434	0.011554	0.008765	0.020319	0.000976	0.008566
DeMarcus Cousins	C	31	718	0.040529	0.006685	0.006964	0.018524	0.000641	0.008078
Anthony Davis	C	28	1404	0.022507	0.002991	0.002564	0.006980	0.000379	0.001994
Dewayne Dedmon	C	32	1065	0.018592	0.002066	0.004601	0.012394	0.000531	0.003099
Andre Drummond	C	28	1437	0.013848	0.003132	0.005498	0.010786	0.000397	0.002714
Andre Drummond	C	28	902	0.018293	0.005987	0.008315	0.018182	0.000596	0.004878
Andre Drummond	C	28	535	0.047850	0.005794	0.015701	0.026168	0.001140	0.006168
Joel Embiid	C	27	2297	0.019634	0.002699	0.001393	0.006138	0.000217	0.002003
Drew Eubanks	C	24	1245	0.017028	0.002651	0.004096	0.007871	0.000479	0.002570
Drew Eubanks	C	24	595	0.030924	0.006723	0.008739	0.017815	0.000887	0.006218
Drew Eubanks	C	24	650	0.036769	0.004000	0.007692	0.013846	0.000994	0.004000

Derrick Favors	C	30	653	0.023583	0.002910	0.006585	0.014395	0.000790	0.001072
Daniel Gafford	C	23	1444	0.016066	0.001593	0.003670	0.005956	0.000480	0.001524
Taj Gibson	C	36	946	0.012791	0.001797	0.004968	0.007928	0.000548	0.001480
Rudy Gobert	C	29	2120	0.011274	0.000802	0.002642	0.007972	0.000336	0.001321
Blake Griffin	C	32	958	0.018998	0.005532	0.003132	0.008873	0.000444	0.001670
Montrezl Harrell	C	28	1641	0.016941	0.002620	0.002681	0.005119	0.000393	0.001280
Montrezl Harrell	C	28	1117	0.025694	0.003850	0.003850	0.008415	0.000577	0.001970
Montrezl Harrell	C	28	524	0.049809	0.008588	0.009351	0.012023	0.001231	0.003817
Isaiah Hartenstein	C	23	1216	0.018668	0.005263	0.003865	0.007155	0.000515	0.002796
Jaxson Hayes	C	21	1398	0.016524	0.001073	0.002933	0.005079	0.000441	0.001359
Willy Hernangómez	C	27	839	0.031824	0.004410	0.010012	0.013707	0.000620	0.003576
Richaun Holmes	C	28	1074	0.019553	0.002048	0.003911	0.009311	0.000615	0.002328
Al Horford	C	35	2005	0.008678	0.002843	0.001347	0.005237	0.000233	0.000798
Dwight Howard	C	36	971	0.018950	0.001751	0.005973	0.012049	0.000630	0.002369
Serge Ibaka	C	32	877	0.023033	0.003079	0.003877	0.011859	0.000570	0.002737
Serge Ibaka	C	32	538	0.039219	0.006134	0.006691	0.018587	0.000911	0.005576
Isaiah Jackson	C	20	541	0.049723	0.001664	0.009982	0.014972	0.001041	0.006654
LeBron James	C	37	2084	0.018714	0.003839	0.000672	0.004367	0.000251	0.002159
Nikola Jokić	C	26	2476	0.016034	0.004685	0.001656	0.006502	0.000235	0.002262
Damian Jones	C	26	1017	0.021141	0.003245	0.003441	0.007965	0.000647	0.002753
DeAndre Jordan	C	33	622	0.025884	0.002572	0.010450	0.022669	0.001034	0.004662
Jock Landale	C	26	589	0.036672	0.006282	0.008659	0.010357	0.000840	0.004244
Alex Len	C	28	620	0.029355	0.005645	0.006452	0.013710	0.000861	0.005323
Kevon Looney	C	25	1732	0.007968	0.002656	0.003406	0.006293	0.000330	0.001097
Robin Lopez	C	33	612	0.032843	0.007026	0.008824	0.007516	0.000904	0.005719
Kevin Love	C	33	1665	0.018138	0.002883	0.001562	0.008108	0.000258	0.001742
JaVale McGee	C	34	1172	0.023805	0.001451	0.005717	0.011689	0.000537	0.003413
Chimezie Metu	C	24	1279	0.015715	0.001798	0.001016	0.008835	0.000353	0.001798
Mike Muscala	C	30	592	0.047804	0.003041	0.002872	0.015034	0.000770	0.002027
Larry Nance Jr.	C	29	858	0.016900	0.004895	0.003846	0.010023	0.000600	0.001748
Nerlens Noel	C	27	562	0.013523	0.003559	0.007651	0.014769	0.000948	0.003203
Jusuf Nurkić	C	27	1579	0.016529	0.003103	0.003293	0.008866	0.000339	0.002787
Onyeka Okongwu	C	21	992	0.019556	0.002621	0.005746	0.008367	0.000696	0.002218
Kelly Olynyk	C	30	764	0.030497	0.009293	0.004450	0.010340	0.000586	0.004843
Mason Plumlee	C	31	1793	0.007139	0.003346	0.002621	0.005800	0.000358	0.001506
Jakob Poeltl	C	26	1970	0.011320	0.002335	0.003249	0.004619	0.000314	0.001371
Bobby Portis	C	26	2028	0.012278	0.000986	0.002120	0.005572	0.000236	0.001036
Dwight Powell	C	30	1798	0.011179	0.001502	0.002670	0.003615	0.000373	0.001001
Naz Reid	C	22	1215	0.020494	0.002305	0.003292	0.006420	0.000402	0.002634
Mitchell Robinson	C	23	1848	0.008929	0.000541	0.004329	0.004708	0.000412	0.000866
Jeremiah Robinson-Earl	C	21	1087	0.015087	0.002024	0.003128	0.008188	0.000381	0.001564
Alperen Şengün	C	19	1489	0.014842	0.003962	0.002955	0.005507	0.000318	0.003089
Isaiah Stewart	C	20	1816	0.008700	0.001211	0.003359	0.005782	0.000281	0.001267
Daniel Theis	C	29	977	0.019447	0.002047	0.002968	0.008598	0.000531	0.002354
Daniel Theis	C	29	584	0.030479	0.002740	0.004452	0.013699	0.000803	0.004281
Karl-Anthony Towns	C	26	2476	0.014095	0.002100	0.001494	0.004120	0.000214	0.001737
Myles Turner	C	25	1235	0.017328	0.001377	0.002105	0.007449	0.000412	0.001700
Jonas Valančiūnas	C	29	2240	0.012946	0.001875	0.002232	0.006071	0.000243	0.001741
Nikola Vučević	C	31	2418	0.010753	0.001985	0.001199	0.005542	0.000196	0.001117
Moritz Wagner	C	24	960	0.029583	0.004583	0.002188	0.010000	0.000518	0.002917
P.J. Washington	C	23	1768	0.010238	0.002262	0.001244	0.003903	0.000266	0.001244
Hassan Whiteside	C	32	1162	0.019535	0.000947	0.006110	0.012048	0.000561	0.001979
Robert Williams	C	24	1804	0.009257	0.001829	0.003659	0.005322	0.000408	0.000942
Christian Wood	C	26	2094	0.013228	0.001671	0.001194	0.006256	0.000239	0.001385
Omer Yurtseven	C	23	706	0.030028	0.004958	0.008499	0.020963	0.000745	0.004108
Ivica Zubac	C	24	1852	0.011231	0.001728	0.003078	0.006102	0.000338	0.001620

6 Cross-Position Discussion

Surprisingly, among these roles, where you play shapes performance more than how old you are. Rebounding? That belongs to centers first, then forwards, leaving guards behind - no matter the years played. Assist numbers jump highest with point guards, pulling well ahead of others across positions. Age changes little; those gaps stay fixed over time.

With advancing years, dispersion patterns shift noticeably across positions. While younger athletes display broader performance ranges - some excelling quickly, others lagging - the data for older ones pulls closer together. This tightening likely emerges as experience shapes consistent contributions over time.

Surprisingly steady scoring rates emerge per minute across age groups when focusing on those with 500 minutes played. That pattern probably appears since the minimum playing time cuts out irregular participants - often younger prospects still growing or aging ones near the edge of squads.

One season's worth of data limits how far conclusions can stretch. Top performers were pulled from a non-random selection, which may tilt results. Player roles within team setups, along with absences due to injury, influence output per minute played. Patterns seen here might hold up under stricter analysis later. Testing them again would help confirm what shows now.

7 Reflection

Surprisingly, diving into this work revealed how crucial stat normalization really is. Without context, raw counts often distort reality - someone logging heavier playing time naturally racks up bigger figures; scaling them down per minute brings things into clearer view.

Putting together three distinct files into a single document raised questions about uniformity. Since each file originally applied its own naming and layout choices, aligning them became essential for meaningful analysis. To manage this, common tools - like `make_panel` and `make_position_figure` - were introduced. Rather than repeat plotting logic across scripts, these functions allowed repetition without redundancy. Consistency improved simply by centralizing what was once scattered.

Putting the code into the appendix helped keep the main section clear. What resulted was a report that flowed better, thanks to separation of details. Readers found it easier to follow when results stayed up front. The structure gave space for analysis without distraction. Length stayed the same, yet clarity improved simply by moving parts around.

Author Contributions

Rishvanth Amsaraj — Forwards analysis (data filtering, normalization, and figure), combining all three position files into the final QMD, patchwork figure layout, shared helper functions, narrative text, and PCIP documentation.

Vishnu Bombesetti — Centers analysis (data filtering, normalization, and figure), and the repository README.md file.

Samarth Puri — Guards analysis (data filtering, normalization, and figure), and the repository PLAN.md file.

The overall analysis approach, filtering criteria, normalization method, and report structure were decided together as a team.

Appendix 1: PCIP Written Plan

The PCIP System (Plan, Code, Implement, Present) was applied to both the data analysis and the coding work in this project.

Phase 1 — Plan

Research question: How does player age relate to six per-minute performance statistics across three NBA position groups (Guards, Forwards, Centers)?

Data source: `nba-player-stats-2021.csv` — 2020–21 NBA season, one row per player.

Issues identified:

- Stats are season totals, so players with more minutes look better without normalization. Dividing by `mp` fixes this.
- Some players have very few minutes, making their per-minute rates unreliable. A 500-minute filter removes these cases.
- The three original files used different variable names and inconsistent formatting, which needed to be unified.

Steps planned:

1. Load `nba-player-stats-2021.csv` once in the setup chunk.
2. Write shared `make_panel()` and `make_position_figure()` helpers to avoid repeating plot code for each position group.
3. For each group: filter by position and `mp > 500`, select the needed columns, divide stats by `mp`, and cap at 100 rows.
4. Produce a 2x3 patchwork figure per group with a shared legend.
5. Write narrative for each section and a cross-position discussion.
6. Set up YAML to match the assignment template (TOC, numbered sections, 1-inch margins, 11pt font, `echo: false` globally).
7. Move all code to a Code Appendix.

Phase 2 — Code

All code follows the **Tidyverse Style Guide** (https://style.tidyverse.org/). Key practices:

- Snake_case for all variable names.
- Pipe operator (`|>`) for data transformation steps.
- Spaces around operators and after commas.
- One `ggplot2` verb per line with consistent indentation.
- Each chunk starts with a comment naming the primary author and reviewer.

Phase 3 — Implement

The code was run in R through Quarto. We checked that filtering produced the right rows for each position group, that all six figures rendered correctly, and that cross-references resolved properly in the PDF.

Phase 4 — Present

The final PDF was produced by rendering `NBA_Age_vs_Performance.qmd`. The report includes a table of contents, three cross-referenced figures with captions and alt text, narrative in every section, a cross-position discussion, a reflection, author contributions, this PCIP plan, and a code appendix.

Appendix 2: GenAI Usage

Tool used: Anthropic Claude Sonnet 4.6 (web interface, claude.ai)

Date of usage: April 28, 2026

Exact prompt (April 28, 2026):

> "I have three 2-by-3 scatter plot figures in my NBA age vs. performance report. Each figure shows six per-minute statistics (points, assists, offensive rebounds, defensive rebounds, field goal percentage, and turnovers) plotted against player age for one position group. The three groups are: Guards (PG and SG), Forwards (PF and SF), and Centers (C). For each figure, write a short fig-alt string (one to two sentences suitable for a screen reader) and a longer plain-English description (three to five sentences) that summarizes the key patterns a sighted reader would take away."

Exact response from Claude (April 28, 2026):

Claude returned the following alt text and long descriptions for each of the three figures. The short alt text strings were used directly in the `fig-alt` chunk options. The long descriptions informed the narrative paragraphs written for Sections 3, 4, and 5.

Figure 1 — Guards (PG and SG)

Short alt text:

> A 2-by-3 grid of scatter plots. The x-axis of every panel shows player age roughly 18 to 38. The y-axes show from left to right and top to bottom: points per minute, assists per minute, offensive rebounds per minute, defensive rebounds per minute, field goal percentage per minute, and turnovers per minute. Points are colored by guard subtype.

Long description:

> Midway through their twenties into their thirties, guards display varied scoring rates per minute, seen clearly in both point and shooting categories. While one might expect sharp differences, the data shows overlap instead. Notably, assists diverge — point guards consistently record higher numbers, matching expectations of them as primary creators. Unlike forwards, these backcourt players rarely collect offensive boards, given their positioning farther from the rim. Among younger cohorts, turnover counts scatter widely, yet they tighten with advancing age, perhaps due to greater experience shaping consistent responsibilities. Efficiency, measured by field goal accuracy, stays mostly flat across years for each group, without distinct peaks or declines.

Figure 2 — Forwards (PF and SF)

Short alt text:

> A 2-by-3 grid of scatter plots. The x-axis of every panel shows player age roughly 18 to 38. The y-axes show from left to right and top to bottom: points per minute, assists per minute, offensive rebounds per minute, defensive rebounds per minute, field goal percentage per minute, and turnovers per minute. Points are colored by forward subtype.

Long description:

Scoring each minute holds steady through the mid-twenties, though a bit more often peaks between ages twenty-six and thirty. While assists generally run lower for forwards compared to guards, several small forwards past age twenty-five post notably higher figures. What stands apart is rebounding — power forwards grab boards far more frequently than small forwards, regardless of age, without narrowing much over time. That pattern hints role shapes rebound output stronger than aging does. Younger players show wider turnover swings early on; these tend to even out later, echoing trends seen earlier in guard data.

Figure 3 — Centers (C)

Short alt text:

> A 2-by-3 grid of scatter plots. The x-axis of every panel shows player age roughly 18 to 38. The y-axes show from left to right and top to bottom: points per minute, assists per minute, offensive rebounds per minute, defensive rebounds per minute, field goal percentage per minute, and turnovers per minute. All points are the same color since all players are Centers.

Long description:

> Scoring rates for centers stay close together — most fall from 0.4 to 0.8 points each minute — with little shift across ages. Because creating shots isn't typically their role, it follows that assist numbers sit below those of other positions. Unlike guards or forwards, these players pull down more rebounds, as seen when comparing across position groups; yet a small decline appears in defensive boards past the early years. By the third decade of life, mobility tends to decrease slightly. Although turnover rates begin elevated among new center players, they tend to decline with experience. This trend mirrors what is observed in backcourt and wing positions alike.

No other GenAI tools were used in this project.

Appendix 3: Code Appendix

```r
# ==========================================================
# Style Guide: Tidyverse Style Guide
# https://style.tidyverse.org/
# ==========================================================

#   Libraries
library(tidyverse)
library(ggplot2)
library(patchwork)
library(knitr)
library(kableExtra)

#   Helper: single scatter panel
# Primary Authors: Rishvanth Amsaraj, Vishnu Bombesetti, Samarth Puri
# Reviewer: All team members
make_panel <- function(df, y_var, y_label, panel_title) {
  ggplot(df, aes(x = age, y = .data[[y_var]], color = pos, shape = pos)) +
    geom_point(alpha = 0.75, size = 1.6) +
    labs(
      title  = panel_title,
      x      = "Age",
      y      = y_label,
      color  = "Position",
      shape  = "Position"
    ) +
    theme_minimal(base_size = 9) +
    theme(
      plot.title      = element_text(hjust = 0.5, size = 9, face = "bold"),
      legend.position = "bottom",
      legend.key.size = unit(0.35, "cm"),
      legend.text     = element_text(size = 7),
      legend.title    = element_text(size = 7)
    )
}

#   Helper: 2x3 patchwork figure for a position group
# Primary Authors: Rishvanth Amsaraj, Vishnu Bombesetti, Samarth Puri
# Reviewer: All team members
make_position_figure <- function(df) {
  p1 <- make_panel(df, "pts",       "Points / Min",        "Points")
  p2 <- make_panel(df, "ast",       "Assists / Min",       "Assists")
  p3 <- make_panel(df, "orb",       "Off. Rebounds / Min", "Off. Rebounds")
  p4 <- make_panel(df, "drb",       "Def. Rebounds / Min", "Def. Rebounds")
  p5 <- make_panel(df, "fgpercent", "FG% / Min",           "Field Goal %")
```

```r
  p6 <- make_panel(df, "tov",       "Turnovers / Min",     "Turnovers")

  ((p1 | p2) / (p3 | p4) / (p5 | p6)) +
    plot_layout(guides = "collect") &
    theme(legend.position = "bottom")
}

# ============================================================
# Section 3: Guards (PG and SG)
# Primary Author: Samarth Puri | Reviewer: Rishvanth Amsaraj
# ============================================================
playerstats <- read.csv("nba-player-stats-2021.csv")

# Check structure and dimensions immediately after import
glimpse(playerstats)

# Filter to guards with > 500 mp; select analysis columns
guard_data <- playerstats |>
  filter(pos == "PG" | pos == "SG") |>
  filter(mp > 500) |>
  select(player, pos, age, mp, pts, ast, orb, drb, fgpercent, tov)

# Normalize each statistic by minutes played using across()
cols_to_normalize <- c("pts", "ast", "orb", "drb", "fgpercent", "tov")
guard_data <- guard_data |>
  mutate(across(all_of(cols_to_normalize), ~ round(. / mp, 6)))
guard_data <- head(guard_data, 100)

make_position_figure(guard_data)

# Summary statistics (mean and SD per variable)
guard_data |>
  select(age, all_of(cols_to_normalize)) |>
  summarise(across(everything(), list(Mean = mean, SD = sd))) |>
  pivot_longer(
    everything(),
    names_to  = c("Variable", ".value"),
    names_sep = "_"
  ) |>
  kable(booktabs = TRUE, digits = 4, linesep = "") |>
  kable_styling(
    latex_options = c("hold_position"),
    font_size = 8
  )

# Full player-level data table
guard_data |>
```

```
  kable(booktabs = TRUE) |>
  kable_styling(
    latex_options = c("scale_down", "hold_position"),
    font_size = 7
  )

# ==========================================================
# Section 4: Forwards (PF and SF)
# Primary Author: Rishvanth Amsaraj | Reviewer: Vishnu Bombesetti
# ==========================================================

# Filter to forwards with > 500 mp; select analysis columns
forward_data <- playerstats |>
  filter(pos == "PF" | pos == "SF") |>
  filter(mp > 500) |>
  select(player, pos, age, mp, pts, ast, orb, drb, fgpercent, tov)

# Normalize each statistic by minutes played using across()
forward_data <- forward_data |>
  mutate(across(all_of(cols_to_normalize), ~ round(. / mp, 6)))
forward_data <- head(forward_data, 100)

make_position_figure(forward_data)

# Summary statistics (mean and SD per variable)
forward_data |>
  select(age, all_of(cols_to_normalize)) |>
  summarise(across(everything(), list(Mean = mean, SD = sd))) |>
  pivot_longer(
    everything(),
    names_to  = c("Variable", ".value"),
    names_sep = "_"
  ) |>
  kable(booktabs = TRUE, digits = 4, linesep = "") |>
  kable_styling(
    latex_options = c("hold_position"),
    font_size = 8
  )

# Full player-level data table
forward_data |>
  kable(booktabs = TRUE) |>
  kable_styling(
    latex_options = c("scale_down", "hold_position"),
    font_size = 7
  )
```

```r
# ==========================================================
# Section 5: Centers (C)
# Primary Author: Vishnu Bombesetti | Reviewer: Rishvanth Amsaraj
# ==========================================================

# Filter to centers with > 500 mp; select analysis columns
center_data <- playerstats |>
  filter(pos == "C") |>
  filter(mp > 500) |>
  select(player, pos, age, mp, pts, ast, orb, drb, fgpercent, tov)

# Normalize each statistic by minutes played using across()
center_data <- center_data |>
  mutate(across(all_of(cols_to_normalize), ~ round(. / mp, 6)))
center_data <- head(center_data, 100)

make_position_figure(center_data)

# Summary statistics (mean and SD per variable)
center_data |>
  select(age, all_of(cols_to_normalize)) |>
  summarise(across(everything(), list(Mean = mean, SD = sd))) |>
  pivot_longer(
    everything(),
    names_to  = c("Variable", ".value"),
    names_sep = "_"
  ) |>
  kable(booktabs = TRUE, digits = 4, linesep = "") |>
  kable_styling(
    latex_options = c("hold_position"),
    font_size = 8
  )

# Full player-level data table
center_data |>
  kable(booktabs = TRUE) |>
  kable_styling(
    latex_options = c("scale_down", "hold_position"),
    font_size = 7
  )
```

References

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Tukey, J. W. (1977). *Exploratory data analysis*. Addison-Wesley.

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